Smaller Diophantine Forms for Con(ZFC)

Two Pell-root translations, joint positivity, and
coefficient-bounded proof certificates

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Abstract

We construct explicit fixed-coefficient Diophantine equations whose lack of solutions is equivalent, by ordinary mathematics formalizable in ZFC, to syntactic Con(ZFC). The principal integer constructions have 26 unrestricted variables at degree 8 and 24 at degree 10; a further restriction of the dyadic coordinate gives 23 at degree 16. The 24-variable equation is a sum of twelve squared residuals. Its reduction uses two integral translations of Pell roots before eliminating coordinates, together with an exact four-witness representation of two strict inequalities. Two positive-domain constructions have 11 variables, degree 277 and 111 arithmetic operations, and 12 variables, degree 78 and 103 operations.

Separately, we prove a new coefficient-budget encoding of finite single-conclusion proof certificates. Products of linear factors recover exact proof-label balances when evaluated at a base exceeding an internally accumulated coefficient bound. This removes the earlier label-difference common-multiple requirement at the source level. We supply its local polynomial and tests, but do not claim a completed fixed-dimensional consistency equation from that architecture. All finished equations retain the cited native-source calibration. No global minimum, unconditional world-first claim, or new complete proof-assistant certificate is asserted.

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1 Results, comparison, and the exact logical claim

The input was the current ProveIt collection `Computability/DiophantineConsistency/con-zfc-diophantine-forms`. The reviewed checkout is pinned to commit

`2605669a532342f153b13eb6b4a0af4b54f5fa37`.

Its latest package reconciliation includes sources 68–75, the tenth wave. The review does not compare with an obsolete earlier wave. The repository already has nine-positive-variable and ten-integer equations of much higher degree, as well as a 38-integer quartic, a 32-integer sextic and a 27-integer octic[1, 4, 5, 6, 7].

There is no agreed total ordering of simplicity. Variable count, domain, degree, circuit size, coefficient height and proof complexity measure different things. The present results improve several precise parts of that comparison; they do not supersede every earlier candidate.

For broader context, the current general representation literature includes Sun's ten-integer result and the parallel formalization work of Bayer, David, Haßler, Matiyasevich and Schleicher[10, 11]. Those universal bounds do not by themselves give a smaller low-degree specialization for this fixed consistency predicate. The positive representation and Pell methods used below also lie in the framework of Sun's earlier work[9].

Construction	Variables	Degree	Operations	Comparison
Integer octic	26	8	–	27-variable degree-8 predecessor
Integer decic	24	10	–	new variable/degree tradeoff
Integer degree-16 companion	23	16	–	26-variable degree-16 predecessor
Positive equation P_{11}	11	277	111	predecessor: 325 / 113
Positive equation P_{12}	12	78	103	predecessor: 82 / 102

An operation is a live binary addition, subtraction or multiplication; squares and multiplication by constants are charged. Fixed integer leaves and reuse of an already computed register are free in this convention. The integer systems are delivered as explicit residual lists and sparse expansions; their evaluator counts have not been optimized or compared. The last positive result trades one operation for four degrees.

1.1 Consistency convention and dependencies

Fix the recursively specified syntactic ZFC proof predicate used by the repository. Write $\mathrm{Ref}_{\mathrm{ZFC}}(p)$ for the statement that p is a finite refutation, and

$$\mathrm{Con(ZFC)} \equiv \neg\exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathrm{ZFC}}(p).$$

The inherited native-source theorems identify the existence of their Diophantine witnesses with the existence of such a refutation. Those theorems include the Wang-tile/native compiler, Pell index reconstruction, and the fixed machine/calculus calibration. We use these as named mathematical inputs and preserve their coefficient conventions.

For each finished equation P with its stated domain D, the result is

$$\mathrm{ZFC} \vdash \left[\neg\exists \boldsymbol{x} \in D^n\ P(\boldsymbol{x}) = 0\right] \iff \mathrm{Con(ZFC)}. \tag{1}$$

The new proofs establish both existence implications without assuming consistency. They then compose with the cited source calibration and negate the equivalence. Thus the argument is neither a bounded search for contradictions nor a proof that ZFC is consistent.

Equation (1) is an ordinary mathematical provability claim. It is stronger than an assertion merely about a few computed witnesses. It is also distinct from delivering a checked term in the repository's internal first-order ZFC calculus. The latter development remains incomplete for these native-source endpoints[8]. This archive adds explicit equations, reductions and exact algebra checks; it does not label finite Python checks as a formal proof of the unbounded semantic calibration.

1.2 What is new in this note

The nearest-square positivity lemma and determinant guard are inherited from different tenth-wave packages. Their composition gives the 26-variable octic. The more consequential algebraic step is to translate *both* Pell roots before removing the c and V coordinates: this yields the 24-variable degree-10 equation while keeping all eliminations integral. The degree-16 companion uses a separately justified divisibility restriction on the canonical dyadic coordinate.

The positive-domain changes likewise combine results that were previously separate: the adjacent-coefficient descent makes a denominator-protection factor redundant, and the adaptive Pell carrier lowers a second degree. The coefficient-budget proof source is a further mechanism, developed here from the previous commutative-flow proposal. Its theorem is complete; its compilation into a competitive whole consistency polynomial is not.

2 Joint guards and two Pell-root translations

This section gives three explicit fixed-coefficient equations over unrestricted integers. The principal constructions have 26 variables and degree eight, and 24 variables and degree ten. A further restriction to canonical dyadic witnesses gives 23 variables and degree sixteen. They improve the inspected repository's 27-variable octic and 26-variable degree-sixteen construction [4, 7]. No global minimum or priority claim outside the inspected repository is made.

The starting point is the literal 38-variable quartic of source 71. Its fixed tuple and determinant guard are kept intact. Two operations are combined here: the simultaneous-positivity lemma of source 68 is applied to the two remaining guards, and two integral translations of Pell-root coordinates permit further substitutions without the usual degree increase. The first operation is a synthesis of earlier results. The paired translations and the resulting 24-variable system are the additional algebraic reduction.

2.1 One guard for two strict inequalities

Put

$$\mathrm{Pos}(p_1, p_2, p_3) = 1 + p_1^2 + p_2^2 + p_3(p_3 + 1).$$

This takes only positive integer values and represents every positive integer. Indeed, apply the three-square theorem to $4n - 3$ and separate its one odd coordinate from its two even coordinates. This gives $4n - 3 = (2p_3 + 1)^2 + (2p_1)^2 + (2p_2)^2$.

Lemma 2.1 (Nearest-square simultaneous positivity; source 68)**.** *For any $x, y \in \mathbb{Z}$,*

$$x > 0 \ \wedge \ y > 0 \quad \Longleftrightarrow \quad \exists t, p_1, p_2, p_3 \in \mathbb{Z} : xy - (t^2 - x - y + 1)^2 = \mathrm{Pos}(p_1, p_2, p_3).$$

Proof. A solution makes $xy > 0$, so x, y have the same nonzero sign. If $x = -a$, $y = -b$, where $a, b \geq 1$, then $(t^2 + a + b + 1)^2 > ab = xy$, a contradiction.

Conversely let $x, y \geq 1$, put $N = x + y - 1$, and take $m = \lfloor \sqrt{N} \rfloor$. Choose $t = m$ if $N - m^2 \leq m$, and $t = m + 1$ otherwise. If N is a square the error is zero. Otherwise $|t^2 - N| \leq m$ and $m^2 < N$. Hence

$$(t^2 - N)^2 < N \leq xy, \qquad xy - N = (x - 1)(y - 1) \geq 0.$$

The remaining positive integer has a Pos representation. The floor and square root are used only to choose a witness in this proof; they are not operations in the polynomial. □

Two separate strict inequalities previously cost six private integer coordinates. Lemma 2.1 uses four. When x and y are quadratic polynomial expressions, its residual has degree at most four. It is essential to keep both inputs quadratic while forming the octic: eliminating ζ too early changes $\eta\zeta$ into a cubic expression.

2.2 Fixed coefficients

The six fixed integers are

$$\beta = 2^{1064921} - 1, \quad K_0, \quad M_C, \quad M_F, \qquad \sigma_F = 2^{1113}, \quad \sigma_C = 2^{666104}.$$

Here (β, K_0, M_C, M_F) is the complete `mixed-parity-unrounded-1064921` tuple of source 34, used without mixing ports from other tuples. In particular M_F is already shifted by β. The archive supplies every actual integer as compressed minimal unsigned little-endian bytes, its SHA-256 digest, and the finite recipe rebuilding it from the literal K215 transition table and ordered 943-tile list. Thus these are fixed numerals, not parameters to be chosen after searching for a solution. The coefficient proof and calibration remain those of sources 34, 64 and 71 [3, 7].

Fixed numeral	Bit length	Minimal raw bytes
β	1,064,921	133,116
K_0	1,065,043	133,131
M_C	1,064,921	133,116
M_F	1,064,922	133,116
σ_F	1,114	140
σ_C	666,105	83,264
Total		615,883

2.3 The octic in 26 integers

Its independent integer coordinates are

$$J, F, \alpha, z, f, h, r, v, w, \tau, \eta, \zeta, y, \gamma, o, j,$$
$$E, c, a, V, I, u, t, p_1, p_2, p_3.$$

The code names v, w, t, p_i as `v_scale`, `w_scale`, `pair_root`, and `guard1`,...,`guard3`. Every expression in the following display is an abbreviation for a polynomial in these 26 coordinates:

$$\begin{aligned}
q &= \beta J + 1, & k &= \eta + \zeta, & Y &= a - E,\\
C &= q - F - \alpha, & X &= 2u^2, & H &= 4a + 3,\\
\delta &= a^2 + H, & T &= E(c - \eta), & S &= X + \gamma H,\\
R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J.
\end{aligned}$$

The two guard expressions and their combined residual are

$$\begin{aligned}
G &= (\sigma_F F)(\sigma_C C) - (q - \sigma_F F - \sigma_C C)^2,\\
Z &= \eta\zeta,\\
B_* &= GZ - (t^2 - G - Z + 1)^2 - \mathrm{Pos}(p_1, p_2, p_3).
\end{aligned}$$

Define the fourteen residuals

$$\begin{aligned}
A_1 &= X - 4qw, & A_2 &= a - E - q^3(2v + 1),\\
A_3 &= E - XY, & A_4 &= c - Yk - \eta,\\
A_5 &= \tau^2 + (2\tau - k)T - 1,\\
A_6 &= 2acS + S^2 - Hc^2 - 1,\\
A_7 &= k - 1 - hE - R,\\
A_8 &= (K_0 + 4w)C - (q-1)z - F,\\
A_9 &= V + c - fo, & A_{10} &= V + k - 1 - cj - hE,\\
A_{11} &= I - (2r+1)c^2, & A_{12} &= I^2 - \delta(f^2 - 1),\\
A_{13} &= I^2(V^2 - y^2) + y^2 - 1, & A_{14} &= B_*.
\end{aligned} \tag{2}$$

The requested single equation is

$$\boxed{Q_{26} = \sum_{i=1}^{14} A_i^2 = 0.} \tag{3}$$

All quantifiers range over $\mathbb{Z}$. The abbreviations introduce no additional quantified coordinates and contain only fixed nonnegative powers.

Proposition 2.2. *The equation $Q_{26} = 0$ has an integer solution exactly when the literal source-71 quartic has an integer solution. Its exact total degree is eight.*

Proof. The source quartic has two rows $G = \mathrm{Pos}$ and $Z = \mathrm{Pos}$, each with its own private triple. Replace these six coordinates and two rows by t, p_1, p_2, p_3 and $B_* = 0$. Lemma 2.1 proves equivalence on the projection to all the other coordinates. This gives a 36-variable system of 24 residuals.

Translate the first root by

$$\tau_{\mathrm{old}} = \tau + E(c - \eta).$$

Then eliminate the following ten monically defined coordinates:

X	$2u^2$	C_2	c^2
F_2	f^2	B_{aux}	I^2
g	$y^2 - V^2$	Δ	$a^2 + 4a + 3$
D	$2u^2 + ac + \gamma(4a + 3)$	P	$k + E(c - \eta)$
L	$q^2 - 1$	N	q^3

These are integer-polynomial restorations with integer-polynomial inverses on the zero sets, given by forgetting the defined coordinates and undoing the root translation. The ten definition rows become identities. For the first and main norms the exact identities are

$$(\tau + T)^2 - (k + T)^2 + (k + T)k - 1 = \tau^2 + (2\tau - k)T - 1,$$
$$(ac + S)^2 - (a^2 + H)c^2 - 1 = 2acS + S^2 - Hc^2 - 1.$$

The surviving rows are exactly (2), up to multiplication of individual residuals by -1, which leaves their squares unchanged. The literal source-row restorations are also checked by exact sparse polynomial arithmetic in the archive.

Every residual has degree at most four. The only residual containing t is B_*, whose t^4 coefficient is -1. Therefore $[t^8]Q_{26} = 1$, independently of the fixed coefficients. This proves exact degree eight on the actual coefficient slice. □

2.4 The degree-ten equation in 24 integers

Retain the same definitions, but now set

$$c = Yk + \eta, \qquad V = fo - c, \qquad y_{\mathrm{old}} = V + \nu. \tag{4}$$

Here c, V are polynomial abbreviations. Replace the coordinate y by ν and remove the coordinates c, V. The 24 independent coordinates are

$$J, F, \alpha, z, f, h, r, v, w, \tau, \eta, \zeta, \nu, \gamma, o, j,$$
$$E, a, I, u, t, p_1, p_2, p_3.$$

Remove A_4, A_9, which now vanish identically. Replace A_{13} by

$$A'_{13} = (V + \nu)^2 - I^2\nu(2V + \nu) - 1. \tag{5}$$

In every other residual substitute the two definitions in (4). The complete single equation is

$$\boxed{\begin{aligned} Q_{24} = {} & A_1^2 + A_2^2 + A_3^2 + A_5^2 + A_6^2 + A_7^2 \\ & + A_8^2 + A_{10}^2 + A_{11}^2 + A_{12}^2 + (A'_{13})^2 + B_*^2 = 0. \end{aligned}} \tag{6}$$

Theorem 2.3 (Two root translations and two coordinate eliminations)**.** *The integer zero sets of Q_{26} and Q_{24} are related by inverse integer-polynomial maps. The polynomial Q_{24} has exactly 24 live integer variables and exact total degree ten.*

Proof. The two definition rows A_4, A_9 are monic in c, V. Their right-hand sides in (4) use only retained coordinates and have degree two. The auxiliary-root translation has the integral inverse $\nu = y_{\mathrm{old}} - V$. The exact identity

$$I^2(V^2 - (V + \nu)^2) + (V + \nu)^2 - 1 = (V + \nu)^2 - I^2\nu(2V + \nu) - 1$$

proves restoration of the auxiliary norm for arbitrary signed integers. There are no sign assumptions or divisibility conditions in these maps.

The degree savings require both root translations. After eliminating c, the expression $T = E(c - \eta)$ has degree three. Centering the first root makes its norm $\tau^2 + (2\tau - k)T - 1$ of degree four instead of six. After eliminating V, the auxiliary-root translation in (5) makes its norm degree five instead of six. In the main norm the cancellation of a^2c^2 already displayed leaves $2acS + S^2 - Hc^2 - 1$, also of degree five. Finally $I - (2r + 1)c^2$ has degree five. All other residuals have degree at most four. In the order displayed in (6), their degrees are

$$2, 4, 3, 4, 5, 4, 2, 3, 5, 4, 5, 4.$$

Thus $\deg Q_{24} \leq 10$.

Only A_{11} contains r. The top part of $c = (a - E)(\eta + \zeta) + \eta$ gives

$$[r^2 a^4 \eta^4] Q_{24} = 4.$$

This coefficient is independent of all six fixed numerals, so the degree is exactly ten after actual coefficient specialization. Exact symbolic checks also verify that every listed coordinate occurs. □

2.5 An optional 23-variable equation of degree sixteen

Starting from Q_{24}, substitute simultaneously

$$u_{\text{old}} = 2q\,u, \qquad w = 2q\,u^2. \tag{7}$$

Then $X = 8q^2u^2$ and $A_1 = X - 4qw$ is identically zero. Remove w from the coordinate list and remove this zero residual. The other eleven squared residuals define $Q_{23,16}$, a polynomial in 23 unrestricted integer variables, with the same six fixed coefficients.

Proposition 2.4. *The absence of integer zeros of $Q_{23,16}$ has the same consistency equivalence as Q_{24}. Its exact degree is sixteen.*

Proof. Any new zero restores a Q_{24} zero by the integral substitutions (7). Completeness uses fresh canonical native witnesses, rather than asserting that every old zero satisfies the restriction. Source 64's canonical witnesses have

$$q = 2^n, \quad n \geq 4, \qquad R \text{ odd}, \quad R > 3q, \qquad u_{\text{old}} = 2^{(R-1)/2}.$$

Since $3 \cdot 2^n > 2n + 3$ for $n \geq 4$, the exponent $(R - 1)/2 - n - 1$ is a nonnegative integer. Thus $2q$ divides u_{old}, and the new u is integral. The old equation $2u_{\text{old}}^2 = 4qw$ then gives exactly $w = 2qu^2$. The source-71 determinant construction retains those canonical coordinates, and the new joint guard and root translations do not change this fact.

With $X = 8q^2u^2$, the main-norm residual has degree eight. Every other residual has degree at most five. Its unique top contribution $X^2 = 64q^4u^4$ gives

$$[J^8 u^8] Q_{23,16} = 4096\beta^8 > 0.$$

Therefore the exact degree is sixteen. This step is an existence equivalence obtained through canonical witnesses, not a bijection of all old zero sets. □

2.6 Consistency theorem and verification boundary

Composing the proved transformations with the named source-71 correctness theorem yields, by ordinary arguments formalizable in ZFC, for each of $Q_{26}, Q_{24}, Q_{23,16}$,

$$\mathrm{ZFC} \vdash \left(\neg\exists \mathbf{x} \in \mathbb{Z}^m \; Q(\mathbf{x}) = 0\right) \iff \mathrm{Con(ZFC)}.$$

The consistency predicate is the specified syntactic ZFC refutation predicate of the inherited calibration. The implication does not assert either consistency or inconsistency of ZFC. No variable coefficient, variable exponent, hidden inequality, or quantified abbreviation is used.

The release checker reconstructs all six actual fixed integers, checks their complete raw hashes, verifies every restored residual exactly, expands each sum of squares over the coefficient ports, proves the reported degree by the displayed nonzero coefficient, and checks variable liveness. Its finite signed and nearest-square samples are only regression diagnostics; Lemma 2.1 is the universal proof. The inherited unbounded native compiler, Pell classification and rank arguments, and literal machine/refutation calibration are not replaced by those finite checks. This package adds no end-to-end proof-assistant or internally encoded ZFC certificate.

3 Two positive-domain improvements by combining the wave-10 mechanisms

The integer-domain construction in the preceding sections is not the only useful endpoint. The same review yields two improvements to the positive frontier. They are separate, fully specified polynomials with the authentic compact44 coefficient tuple. Their contribution is a precise composition of the mechanisms in sources 69 and 70, rather than a claim that the generic norm or descent methods are new.

Construction	Positive variables	Exact degree	Operations
Source 70	11	325	113
Guard-free strong ordinate	11	277	111
Source 69	12	82	102
Adaptive main carrier	12	78	103

The operation convention counts every live binary addition, subtraction, and multiplication on witnesses, including squaring and scalar multiplication. Fixed integer leaves and register reuse are free. These are arithmetic-circuit counts, not bit-complexity estimates.

3.1 An adjacent-coefficient descent

We use the following arithmetic lemma from source 69[5]. Its role here is to remove a hypothesis that source 70 enforced by an additional factor in its strong Pell ordinate.

Lemma 3.1 (Source 69: rational ordinate descent)**.** *Let $Q > 1$ and V be integers. If $y \in \mathbb{Q}$ and*

$$QV^2 - (Q-1)y^2 = 1,$$

then $y \in \mathbb{Z}$.

Proof. Write $y = a/m$ in lowest terms, with $m > 0$. The equation implies $m^2 \mid Q - 1$, and excludes $V = 0$ and $y = 0$. Change signs to make both positive. If $V = 1$ then $y = 1$. Otherwise define

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV.$$

The first is integral because $m \mid Q-1$. Direct expansion preserves the norm. The reduced denominator of y' is exactly m, since $2Q-1 \equiv 1 \pmod{m}$. Moreover $y > V$ and $QV > (Q-1)y$, and hence $-V < V' < V$. The norm excludes $V' = 0$. Strong induction on the positive integer V, after taking absolute values, reaches the case $V = 1$ without changing the denominator. Thus $m = 1$. □

The exact polynomial norm identity in this proof is checked by the supplied twelve-variable verifier. No finite enumeration is being used in place of the induction.

3.2 Eleven positive unknowns in degree 277

Use the eleven inputs and every register of source 70's eleven-positive construction[6], with exactly one change:

$$\boxed{I = ic^2 \quad \text{in place of} \quad I = iq(q-1)c^2.}$$

The inputs are

$$(j, F, \alpha, i, T, X, Y, n, \eta, \zeta, \gamma) \in \mathbb{Z}_{>0}^{11}.$$

The fixed numerals are β, K_0, M_C, M_F, with $\beta = 2^{590069} - 1$ and the already-shifted M_F. All abbreviations below are polynomial registers, not extra unknowns. For completeness the definition is

$$\begin{aligned}
J &= 3j+1, & q &= \beta J + 1, & C &= q - F - \alpha,\\
E &= XY, & k &= \eta + \zeta, & c &= kY + \eta,\\
a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H,\\
\ell &= EkY, & I &= ic^2, & M &= q^3(q-1),\\
R &= ((q-1)(q-F)+\alpha)(q^2-1) + (M_C + qM_F)J, & & & &\\
R_+ &= q^3(q+F+\alpha), & B_t &= (qK_0 + X)C - qF, & D_\tau &= \ell(\ell+k)+1,\\
D_f &= 1 + \Delta I^2, & Q &= \Delta(D_f - 1), & \delta &= Q - 1,\\
V &= D_f(cT - R) - c, & L &= 3D_f(cT + R_+), & W &= 1 + (q-1)L,\\
Z &= M(\ell + k + n + L) + (q-1)Y - q^3 W B_t, & & & &\\
S &= 4qY, & U &= qS, & v &= k - R - 1,\\
D_* &= U(X + ac) + H(U\gamma + SX - v), & & & &\\
A_1 &= D_*^2 - U^2(1 + \Delta c^2). & & & &
\end{aligned}$$

Now put

$$\begin{aligned}
A &= M^2 D_\tau, & B_f &= M^2 D_f, \quad C_y = M^2(QV^2 - 1),\\
N_c &= \delta^2\left[(Z^2 - A - B_f)^2 - 4AB_f\right] - 2\delta C_y(Z^2 + A + B_f) + C_y^2,\\
N_8 &= N_c^2 - 64\delta^3 A B_f C_y Z^2.
\end{aligned}$$

The required equation is

$$\boxed{P_{11} := (1 + N_8)(1 + A_1) - 1 = 0.} \tag{8}$$

The exact division-free definition is also supplied as `positive/eleven/p11_positive_degree277.json`.

Proposition 3.2. *ZFC proves that (8) has no zero in $\mathbb{Z}_{>0}^{11}$ if and only if* Con(ZFC)*. Its exact total degree is 277 and its emitted evaluator has 111 live binary operations.*

Proof. We give the complete change to the upstream proof; its normalized Pell and native compiler interfaces remain the ones in sources 61 and 70 [2, 6]. The source 70 inequalities use only $I \geq c^2$, which still holds. In particular, writing

$$D_y = \frac{QV^2 - 1}{Q - 1},$$

the three positive rational radicands D_τ, D_f, D_y are pairwise distinct, and no signed sum of their square roots is zero. The octic norm therefore recovers rational signed roots τ_*, f_*, y_* with squares D_τ, D_f, D_y. The first two are integers because their squares are integers. Lemma 3.1 makes the third an integer. Here Q, V are integers and $Q > 1$ before any equation is imposed.

This replaces the old argument that the denominator of y_* divides M and is coprime to $Q - 1$. That argument was the only reason for the extra $q(q-1)$ in I. The unchanged recovery formulas are

$$f_{\text{old}} = |f_*|, \quad y_{\text{old}} = |y_*|, \quad u = y_{\text{old}} - V > 0, \quad r = n + L - f_* - y_* > 0.$$

The signed roots must be used in the formula for r. The source proof's bound $L > |f_*| + |y_*|$ makes it positive. The strong ordinate $I = ic^2$ has exactly the divisibility needed by the normalized index theorem. Consequently the original four-residual positive source interface is recovered, and its calibration yields a finite syntactic ZFC refutation.

Conversely any positive zero of source 70 maps directly to a zero here: replace its input i by $q(q-1)i$ and leave every other input unchanged. This preserves I and every downstream register. Hence source 70's completeness supplies completeness here with no new index restriction. The unit-product finalizer is unchanged: both N_8 and A_1 are squares modulo 4, so neither can equal -2.

For exact degree, the changed register degrees are

register	I	D_f	Q	L	W	Z	N_c	N_8	A_1
degree	5	14	18	18	19	24	132	264	13.

Writing $q_0 = 3\beta j$, $k_0 = \eta + \zeta$ and $B_0 = (K_0 q_0 + X)(q_0 - F - \alpha) - q_0 F$, the leading forms are

$$\begin{aligned} Q_0 &= i^2 k_0^4 X^4 Y^8, \\ Z_0 &= -3i^2 q_0^7 k_0^4 X^2 Y^6 (q_0 + F + \alpha) B_0, \\ (A_1)_{13} &= 32 q_0^2 k_0 X^2 Y^4 \big(4 q_0^2 Y \gamma + q_0^3 (q_0 - F)\big). \end{aligned}$$

The unique top term of N_8 is $Q_0^4 Z_0^8$. Therefore the coefficient of

$$j^{60} i^{24} \eta^{49} X^{42} Y^{85} \alpha^{16} \gamma$$

in P_{11} is $2^7 3^8 (3\beta)^{60} > 0$. The monomial has degree 277. The exact checker verifies these identities and the literal graph count: 65 multiplications, 29 additions and 17 subtractions. □

3.3 Twelve positive unknowns in degree 78

For a second endpoint, keep all source 69 twelve-variable registers [5], except replace its main Pell carrier by

$$\begin{aligned} S &= 4qY, & U &= qS = 4q^2 Y, \\ D_* &= U(X + ac) + H\big(U\gamma + SX - (k - R - 1)\big), \\ A_1 &= D_*^2 - U^2(1 + \Delta c^2). \end{aligned}$$

The source's first carrier A_0, two-root resultant N, and finalizer are unchanged:

$$\boxed{P_{12} = (1 + A_0)(1 + A_1)(1 + N) - 1 = 0.} \tag{9}$$

Its full literal definition is `positive/twelve/equation.json`; the twelve positive inputs are $j, F, \alpha, t, T, X, Y, r, \eta, \zeta, n, \gamma$.

To make the unchanged part explicit, use $J = 6j + 1$, $q = \beta J + 1$, $C = q - F - \alpha$, and the same $E, k, c, a, H, \Delta, \ell, R, B_t$ formulas as above. Put

$$\begin{aligned}
W_0 &= 1 + (q-1)k, & M &= q^3(q-1),\\
Z_0 &= M(\ell + k + r) + (q-1)Y - q^3 W_0 B_t,\\
A_0 &= Z_0^2 - M^2(\ell(\ell+k)+1),\\
f &= t + 1, & \rho &= f^2 - 1, & d &= \Delta c^4,\\
Q &= \Delta\rho, & V &= f^2(cT - R) - c,\\
h(z) &= (1-Q)z^2 + 2(1-Q)Vz + V^2 - 1,\\
A_N &= dh(n) + (1-Q)\rho, & B_N &= 2(1-Q)(n+V),\\
N &= A_N^2 - dB_N^2\rho.
\end{aligned}$$

All coefficients are fixed integers and all powers have fixed exponents.

Proposition 3.3. *ZFC proves that (9) has no zero in $\mathbb{Z}^{12}_{>0}$ if and only if* Con(ZFC)*. Its exact total degree is 78 and its emitted evaluator has 103 live binary operations.*

Proof. The unit-product finalizer still works: q is even, $4 \mid U$, and all three residuals are squares modulo 4. First $A_0 = 0$ gives $q^3 \mid Y$ and $C > 0$, hence $q \geq 16$, $q \equiv 2 \pmod 3$ and $3q < R < q^4 - 2$. The new D_* is positive, since $D_* > Uac - Hk > ak(UY - 5) > 0$. Thus $A_1 = 0$ recovers the positive main Pell root. The first and main roots give $c > 2R$ before the strong or auxiliary equations are used. Consequently $V > 1$, and the unchanged resultant and Lemma 3.1 recover positive integral strong and auxiliary coordinates. The normalized auxiliary-index theorem then gives $p = R$.

For the altered carrier set $\mu = S/\gcd(S, H)$. Since $q^3 \mid Y$, $q \equiv 2 \pmod 3$ and $H = 4Y(X+1) + 3$, one has $\gcd(S, H) \leq 3$. Thus $\mu \geq 4qY/3 > q^4 > R + 1$. Both S and U divide $4XY^2$ without assuming $q \mid X$. Carrier divisibility gives $\mu \mid k - R - 1$, and the first Pell recurrence gives $k \equiv 2n_{\text{first}} \pmod{4XY^2}$. With $1 \leq n_{\text{first}} < p = R < \mu$, these imply $2n_{\text{first}} = R + 1$. Therefore $v = k - R - 1$ is a positive multiple of $4XY^2$, hence of U. Returning to the carrier gives $q \mid HX$, and $\gcd(q, H) = 1$ gives $q \mid X$.

Writing $w = X/q$, the restored source gap is $\gamma_{\text{old}} = \gamma + w - v/U$. It is integral and positive because the main root is greater than $(a+1)c$ and $c > X$. This recovers the source 69 interface. Conversely its canonical witnesses lift by $\gamma = \gamma_{\text{old}} + v/U - w > 0$, using $v/U \geq XY/q^2 = wY/q \geq wq^2 > w$. No new canonical restriction is introduced.

The exact residual degrees are $15, 13, 50$. Their product has degree 78: its coefficient at

$$j^{18}F^2\eta^{11}X^{12}Y^{23}t^8n^2\gamma\alpha$$

is $1024(6\beta)^{18} > 0$. The supplied checker verifies the complete carrier expansions, resultant identities and this coefficient. It counts 58 multiplications, 26 additions and 19 subtractions. Thus this endpoint trades one operation for four degrees compared with source 69. □

These two improvements preserve the native compiler and syntactic consistency calibration. Their exact algebra is independently replayable, but they are not presented as new end-to-end proof-assistant certificates.

4 Proof certificates with linear factors and a coefficient budget

This section develops a different arithmetic source for proof existence. The proved contribution is a certificate theorem and its literal local polynomial. The final elimination of the unbounded trace has

not been materialized or measured. Thus the local counts below are not a claim of a new complete Diophantine-variable record.

Source 68 of the repository[4] already replaces ordered proofs by nonnegative Horn flow and encodes formula labels with the pairwise coprime numbers $1 + Mj$. It requires a second bounded universal block ensuring that M is divisible by all integers up to a label bound. We remove that requirement. We use the linear factors $X + j$, respect the actual arity of each rule, and carry a separate, small arithmetic budget bounding polynomial coefficients. The budget is chosen before the evaluation point. No factorial, primality test, common-multiple test, or variable exponent occurs in the new source. The later elimination of bounded universal quantification may still use such graphs; their costs have not disappeared from a final compiler by assertion.

4.1 Exact evaluation detects a polynomial identity

All formula labels are natural numbers, including zero. A rule has one conclusion and zero, one, or two premises. An occurrence of zero as an actual premise is a genuine obligation. In the C222 application, zero is discharged by the explicit truth axiom; it is not discarded as a free coordinate in the following construction.

For a finite list of rule instances $R_i = (P_i \longrightarrow c_i)$, $0 \le i < N$, put

$$B_i = \prod_{p \in P_i} (1 + p) + c_i, \qquad T_0 = 1, \qquad T_{i+1} = T_i B_i, \tag{10}$$

$$C(X) = \prod_{i<N} (X + c_i), \qquad Q(X) = \prod_{i<N} \prod_{p \in P_i} (X + p), \qquad M = T_N + 1. \tag{11}$$

Products count repeated premises with their multiplicities. An empty product is one. If two numerical premise slots p_i, q_i are used, pad unused slots with zero. Then

$$B_i = (1 + p_i)(1 + q_i) + c_i$$

for a binary rule; the same expression specializes to $1 + p_i + c_i$ for a unary rule and to $1 + c_i$ for an axiom.

Lemma 4.1 (Coefficient budget). Every coefficient of $C(X) - (X + 1)Q(X)$ has absolute value at most T_N. Consequently,

$$C(M) = (M + 1)Q(M) \quad \Longleftrightarrow \quad C(X) = (X + 1)Q(X) \text{ in } \mathbb{Z}[X]. \tag{12}$$

Proof. Each B_i is at least both $1 + c_i$ and $\prod_{p \in P_i}(1 + p)$. Thus $C(1) \le T_N$ and $Q(1) \le T_N$. Write $Q(X) = \sum_j q_j X^j$, with $q_j \ge 0$ and missing coefficients zero. The coefficient of X^j in $(X + 1)Q(X)$ is $q_j + q_{j-1}$. These are distinct entries in the coefficient list, so their sum is at most $Q(1)$, including at either endpoint. Every coefficient of C is at most $C(1)$. The difference of two nonnegative numbers each at most T_N has absolute value at most T_N, not $2T_N$.

It remains to use a simple injectivity argument. If a nonzero integer polynomial $H(X) = \sum h_j X^j$ has $|h_j| < M$, let k be the least index with $h_k \neq 0$. Dividing $H(M) = 0$ by M^k and reducing modulo M gives $M \mid h_k$, contradicting $0 < |h_k| < M$. Apply this to $H = C - (X + 1)Q$ and $M = T_N + 1$. The reverse implication is immediate. □

The factor two sometimes used in such a coefficient bound is unnecessary. Also, the product $\prod_i (1 + c_i)(1 + p_i)(1 + q_i)$ is an unnecessarily large budget. The factor B_i in (10) gives the required two majorizations separately and lowers the budget update from degree four to degree three in its displayed coordinates.

Lemma 4.2 (Linear-factor uniqueness). The polynomial identity in (12) holds exactly when

$$\sum_{i<N}\left(e_{c_i}-\sum_{p\in P_i}e_p\right)=e_1 \quad \text{in the free abelian group on } \mathbb{N}. \tag{13}$$

Proof. The distinct monic linear factors $X+j$, $j\in\mathbb{N}$, have distinct roots. Their multiplicities in an identity of products are equal. This can be proved without invoking a general factorization theorem: cancel the common copies of $X+j$ from the two sides and evaluate at $X=-j$; a remaining copy on only one side would make exactly that side zero. This applies also to $j=0$, whose factor is X. The resulting multiplicity equations are precisely (13). □

4.2 Why cycles cannot manufacture a refutation

Let $\mathcal{R}$ be any collection of single-conclusion rules with at most two premises and labels in $\mathbb{N}$. Let D be its least set of derivable labels, starting from the zero-premise rules.

Lemma 4.3 (Single-conclusion flow). Label 1 is derivable if and only if some finite list of instances from $\mathcal{R}$, with repetitions permitted, satisfies (13).

Proof. Unroll a finite proof into a tree. Every non-root node contributes once as a conclusion and once as a premise of its parent, leaving only the root vector e_1.

Conversely, for a proposed finite list, form the finite closure of just the rules in that list. If 1 is outside this closure, sum the coordinates of the alleged flow equation that correspond to labels outside the closure. Whenever a conclusion is outside, at least one premise is outside, since the set is closed under the listed rule. The contribution of that rule to this sum is therefore at most zero. When its conclusion is inside, its contribution is again at most zero. The total on the left is at most zero, whereas the right side contributes one. This is impossible. The finite closure also gives a constructive proof DAG, which can then be unrolled to a finite tree. □

Repeated premises, self-loops, and disconnected cyclic components do not invalidate the proof. The single-conclusion condition is essential: a rule $a\longrightarrow(a,1)$ would have net vector e_1 without a way to derive either conclusion from no axioms.

Theorem 4.4 (Two-accumulator certificate). Label 1 is derivable in $\mathcal{R}$ if and only if there exist N,M, a list of N valid rule instances $P_i\longrightarrow c_i$, and natural sequences $S_0,\dots,S_N$, $T_0,\dots,T_N$ such that

$$S_0=M+1, \qquad S_N=1, \qquad T_0=1, \qquad M=T_N+1, \tag{14}$$

$$S_{i+1}(M+c_i)=S_i\prod_{p\in P_i}(M+p), \qquad (0\le i<N),$$

$$T_{i+1}=T_i\left(\prod_{p\in P_i}(1+p)+c_i\right), \qquad (0\le i<N). \tag{15}$$

No external bound on the labels is required.

Proof. For soundness, the budget recurrence implies $T_i\ge 1$ and hence $M\ge 2$. Every factor $M+j$ is positive. Starting from $S_0=M+1>0$, the first recurrence shows inductively that every S_i is positive. Multiply all of these recurrences and cancel the nonzero internal accumulators. The endpoints give $C(M)=(M+1)Q(M)$. The coefficient-budget lemma, linear-factor uniqueness, and the flow lemma now prove derivability of 1.

For completeness, start with a finite proof tree. List its nodes in a top-down order in which the next expanded node is on the current frontier. First compute the entire sequence T_i from this list of

labels; set $M = T_N + 1$ only afterwards. Let S_i be the product of $M + j$ over the open frontier at stage i. Initially the frontier is just label 1; at the end it is empty. Expanding a node labelled c_i removes its factor $M + c_i$ and inserts the factors for its actual premises. This gives both recurrences and all four endpoints. A zero-labelled leaf is expanded with its actual zero-premise truth axiom when that axiom is used. There is no circular choice of a scale or upper bound. □

The empty list cannot certify target 1: it would require simultaneously $M = 2$ and $S_0 = S_N = 1 = M + 1$. The argument is entirely finitary and uses ordinary induction, integer arithmetic, and a finite closure construction. It can be carried out within ZFC, without a standard-model soundness assumption or an assumption that ZFC is consistent.

4.3 The local arithmetic rows are quadratic after branch sharing

Write the current and next accumulator values as S, S', T, T', and use p, q, c for the two premise slots and conclusion. The local flow and budget constraints are as follows:

Rule arity	Flow equation	Budget equation
Zero	$S'(M + c) = S$	$T' = T(1 + c)$
One	$S'(M + c) = S(M + p)$	$T' = T(1 + p + c)$
Two	$S'(M + c) = S(M + p)(M + q)$	$T' = T(1 + p + q + pq + c)$

For the explicit truth axiom $c = 0$, the first row specializes further to $S'M = S$ and $T' = T$.

Only the binary row needs additional local arithmetic wires. Introduce

$$u = S(M + p), \qquad w = pq.$$

Then its four residuals are

$$u - S(M + p), \quad S'(M + c) - u(M + q), \quad w - pq, \quad T' - T(1 + p + q + w + c). \tag{16}$$

They are all quadratic. The witnesses u, w are uniquely determined and natural whenever the displayed input coordinates are natural.

The supplied local JSON augments the literal branch rows of source 68's `horn_shared_source.json`. Its binary modus-ponens branch originally uses no scratch wires, so u, w occupy its existing shared slots `q0`, `q1`. No active branch loses a wire or shares two distinct values in one slot. The most demanding inherited branch still uses thirteen slots. The generalization branch retains its own original `q0`; it needs no new arithmetic wire. The ten zero-premise branches, including truth, need no new arithmetic wire either.

For each branch, add the displayed residual squares to that branch's existing sum of squares. Multiply the twelve resulting nonnegative branch polynomials. Over the naturals it vanishes exactly when one valid branch and its corresponding accumulator update hold. The construction has twenty local existential coordinates:

$$p, q, c, \quad a, b, c_{\text{schema}}, d, \quad q_0, \ldots, q_{12},$$

where the JSON uses `v` for the conclusion to avoid the displayed name collision. Its five exposed coordinates are S, S', T, T', M. The truth factor rises from degree two to degree four, the inherited bundle factor stays at degree eight, and all other factors have degree four. The local polynomial therefore has exact degree $11 \cdot 4 + 8 = 52$. This follows both from exact sparse expansion of every factor and from nonvanishing of products over $\mathbb{Z}$.

The following comparison uses exactly the same inherited Horn rows and the same unflattened branch-product representation. It does not compare the size of two final Diophantine polynomials.

Source-interface quantity	Source 68	This construction
Local existential coordinates	23	20
Exposed local coordinates	4	5
Local polynomial degree	50	52
Outer certificate coordinates	5	4
Bounded universal families	2	1
Beta values needed at one step	2	4

The extra beta values are a real cost: they encode the second accumulator. Their arithmetic graph witnesses must be counted before asserting a net improvement in a compiled polynomial.

4.4 A completely specified trace interface

Use four outer natural coordinates N, x, y, M and

$$\beta(x, y, j) = x \bmod (1 + (j + 1)y), \qquad S_i = \beta(x, y, 2i), \qquad T_i = \beta(x, y, 2i + 1).$$

Impose the four endpoint equations (14). For every $i < N$, require a zero of the supplied local polynomial with exposed coordinates $(S_i, S_{i+1}, T_i, T_{i+1}, M)$. This is one bounded universal family; there is no additional family expressing divisibility of a scale by all labels.

Every finite pair of accumulator sequences has such a common beta code. For example, choose y larger than all sequence entries and divisible by $1, 2, \ldots, 2N+1$. The moduli $1+(j+1)y$, $0 \le j \le 2N+1$, are pairwise coprime: a common divisor is coprime to y, divides the index difference, and that difference divides y. The Chinese remainder theorem supplies x, and the choice of y makes the desired residues their least nonnegative remainders. This choice is made after M and the two traces; it creates no circular scale dependency.

For the calibrated C222 rule relation, the resulting finite-certificate predicate is equivalent to existence of a C222 refutation of label 1. Composing with that calculus's inherited syntactic calibration gives the same consistency target. This route checks rule instances directly; its source does not simulate K215 and does not use a Jones finalizer.

The remaining work is precise. Substitute literal remainder graphs for the four beta values, apply a certified bounded-universal elimination to the unflattened local source, quantify the four outer coordinates, and only then choose a final degree-reduction or variable-compression strategy. The full ordinary polynomial, its complete variable count, and its coefficient payload have not been produced here. The theorem above is not a substitute for those artifacts.

4.5 An optional normalized variant

If one wishes to retain the earlier free-zero convention, use weights $1 + jX$ and pad every rule to two premises. Keep $T' = T((1 + p)(1 + q) + c)$ and $M = T_N + 1$. The same coefficient proof applies to products of $1 + jX$, and factor uniqueness recovers all positive-label multiplicities; the factor for zero is one. The soundness theorem then uses zero as a designated derivable label.

The flow equation is

$$S'(1 + Mc) = S(1 + Mp)(1 + Mq).$$

It has a useful exact quadratic lift with four natural auxiliary values:

$$w = pq, \quad u = MS, \quad u' = MS', \quad v = Mw.$$

The six quadratic residuals are

$$w - pq, \quad u - MS, \quad u' - MS', \quad v - Mw,$$
$$S' + cu' - S - u(p + q + v), \qquad T' - T(1 + p + q + w + c).$$

Their sum of squares is quartic, and its zeros are exactly the two desired updates together with the uniquely determined natural auxiliaries. Expanding the fifth residual gives $S'(1+Mc)-S(1+Mp)(1+Mq)$ exactly. This optional construction explains how to reuse the previous padded Horn-flow interface, but the unnormalized linear-factor source is cheaper locally because it respects rule arity.

4.6 Checks and a necessary negative control

The standalone checker reconstructs all twelve branch sum-of-squares polynomials exactly from their literal residuals. It checks 178 exact natural branch fixtures against independently written C222 constructor formulas, including both signs of the guarded bundle. It exhausts all 60,880 lists of up to three zero-, one-, or two-premise rules on labels $\{0, 1, 2\}$, checking the coefficient bound, numerical identity, polynomial identity, flow equation, and derived-target implication. Four toy proof-tree fixtures cover repeated premises and explicit truth leaves; they are finite artificial Horn systems, not ZFC refutations. The optional normalized lift has its own exact arithmetic checks.

The budget endpoint is essential. Take $M = 11$ and the two rules $10 \longrightarrow 3$ and the axiom 7. The target 1 is underivable, but

$$(M + 3)(M + 7) = (M + 1)(M + 10), \qquad 14 \cdot 18 = 12 \cdot 21.$$

Indeed $S = (12, 18, 1)$ satisfies both local flow updates, and M is larger than every label. The budget is $T = (1, 14, 112)$, so the required endpoint would be $M = 113$, not 11. The complete checker rejects the fake. Thus an unspecified assertion that the evaluation point is "large" cannot replace the coefficient proof.

These finite checks are diagnostic evidence for the implementation. The general mathematical justification is the preceding proof, and no kernel-checked formalization of this new source is claimed.

5 Coefficient specification and reproducibility

The display letters for fixed coefficients are abbreviations for actual integers, not additional quantified unknowns and not a choice depending on the truth of consistency. The archive contains their complete compressed little-endian byte representations, exact sizes and SHA-256 digests. The integer component also reconstructs its coefficient tuple from the supplied 943-tile list and fixed machine table and compares every byte.

The integer equations use the whole `mixed-parity-unrounded-1064921` tuple: $\beta = 2^{1064921} - 1$, the inherited K_0, M_C, M_F, and the determinant scales 2^{1113} and 2^{666104}. Their six minimal unsigned payloads total 615,883 raw bytes. The positive equations use the different whole compact44 tuple, with $\beta = 2^{590069} - 1$. The two tuples are never intermixed. In each, M_F is already the shifted source mask; applying a second shift would specify a different equation.

5.1 Files that define the equations

The integer packets under `integer/generated/` give the ordered unknowns, residuals, fully expanded abbreviations and source restoration maps. For a packet with residuals $r_1, \ldots, r_s$, its single equation is exactly $\sum_{i=1}^{s} r_i^2 = 0$. Sparse expanded polynomial files are provided as additional representations over the fixed coefficient ports. Substituting the supplied fixed integers gives the required element of $\mathbb{Z}[x_1, \ldots, x_n]$.

The positive packets are finite straight-line programs in the three binary operations $+, -, \times$. Their register names introduce no additional existential variables. Every operation is listed, and a designated final register defines the polynomial. There is no variable exponent, implicit division, unbounded product or uninterpreted predicate in any finished equation.

5.2 Checks and their mathematical scope

Run `python3 verify.py` at the archive root. The normal verification path uses the Python standard library. It checks authentic payloads, regeneration of all equation packets, exact polynomial identities, residual restoration, variable liveness, and the degree witnesses stated in the proofs. Optimized Python runs of the components were also checked; the checks do not disappear when `assert` is disabled.

The additional `independent_integer_audit.py` uses SymPy and does not import the construction program or its polynomial engine. It independently restores every row of the joint-guard source and checks the exact lower-bound coefficients. An independent modular univariate specialization of the positive circuits supplies another check on their literal degree. These checks address transcription and algebra. The universal existence arguments are the written proofs, and the inherited native-source semantic theorems remain explicitly cited inputs.

The direct component includes positive proof-tree examples, repeated and zero-premise cases, cycle controls, and a false large-base product collision. They test the source implementation. The general soundness argument uses coefficient bounds and finite Horn closure, not successful examples.

For the document, run `pdflatex report.tex` twice or use the supplied `Makefile`. The archive carries source pins, notices and licenses for all reused material. It includes a content manifest; intermediate LaTeX files and Python caches are excluded from the release archive.

6 Further research questions

1. **Can the coefficient-budget source be compiled efficiently?** The next decisive deliverable is a sharing-preserving arithmetic representation of its single bounded family of local checks, followed by a measured fixed-coefficient polynomial. The local witness count is not the final witness count. The generic bounded-universal compiler may reintroduce factorial or power representations internally; the present source theorem removes its own extra common-multiple block only.

2. **Can 24 unrestricted variables be reached at degree 8?** The remaining degree-5 residuals in the decic have explicit origins: the centered main norm, the strong ordinate, and the centered auxiliary norm. A further integral coordinate change must control all three at once. Reducing a formal degree before a required substitution is not enough.

3. **Can the 23-variable companion have degree below 16?** Its degree comes from the square of the new dyadic term in the main residual. Another order of elimination or a different coordinate used to carry the dyadic divisibility could reduce it. Any restriction must have a proved lift from authentic canonical source witnesses.

4. **Can four Pell radicals replace another carrier variable?** Appendix A proves an unconditional separation fact for four radicals in the positive construction. The open difficulty is preserving two carrier congruences with one integer coordinate. Their denominators share powers of q, so a naive rational-sum merger allows cancellation and does not prove the desired divisibilities.

5. **Can the native congruence move to a different root?** Supplying F, α, C directly makes positivity cheap, but the constraint $q \equiv \beta + 1 \pmod{3\beta}$ still needs to be enforced. Inserting it into the first root's rational offset can corrupt the transport congruence when denominators share factors. A useful solution would preserve the integer strong-root coordinate and allow the two-root descent to apply.

6. **Can the whole proof become internally checked?** The latest repository already contains a closed numeric sealed-ZFC refutation checker. Its internal interpretation and the full native compiler calibration are separate formalization projects. The finite algebraic lemmas here are

suitable small targets; they should not be confused with the much larger final consistency biconditional.

A A four-radical separation lemma for a possible next merger

This appendix records a proved ingredient for the open direction above. It does not define a new ten-positive-variable consistency equation. Use the eleven-positive construction's registers, with $I = ic^2$, and set

$$t = \sqrt{\ell(\ell+k)+1}, \quad g = \sqrt{1+\Delta c^2}, \quad f = \sqrt{1+\Delta I^2}, \quad y = \sqrt{\frac{QV^2-1}{Q-1}}.$$

These are real positive roots for every positive input, before imposing any polynomial equation. Let $b = cT - R$, so $V = f^2 b - c$.

Lemma A.1. *Every nonempty signed sum of a subset of t, g, f, y is nonzero. Consequently, if a signed sum of all four is rational, then all four roots are rational.*

Proof. Here $a \geq 2$, $c \geq 3$, $I \geq c^2$ and

$$t < \ell + k < (a+1)c < g < (a+2)c, \qquad f > (a+1)c^2 > g + t.$$

As in the three-root construction, $|V| > 1$ and $y^2 = V^2 + (V^2-1)/(Q-1)$. The three integer radicands for t, g, f are strictly increasing and less than Q. If y^2 equalled any one of them, say d, then $Q(d - V^2) = d - 1$ with $0 < d - 1 < Q$, impossible. Thus all four roots are pairwise distinct.

If $b \neq 0$, then $y > |V| \geq f^2 - c > 2f > f + g + t$. Together with $f > g + t$ and $g > t$, this prevents every signed cancellation. If $b = 0$, then $V = -c$ and $Q > c^2$, so $y < c + 1$. Moreover

$$(a+1)c - (\ell + k) - (c+1) = k(Y^2 - 1) + a\eta - 1 > 0,$$

which gives $g > t + y$. Finally

$$g + t + y < (2a+4)c + 1 < (a+1)c^2 < f.$$

The last strict inequality in the middle follows from $a \geq 2, c \geq 3$. Thus a subset containing f cannot cancel; without f, a subset containing g cannot cancel; and the remaining two distinct positive roots t, y cannot cancel.

For rationality, work in the finite multiquadratic field generated by the four roots. A rational signed sum is fixed by every automorphism. Subtract its image under any automorphism. If that automorphism changes any root's sign, the difference is twice a nonempty signed subset sum, which has just been excluded. Every automorphism therefore fixes every root. The fixed field is $\mathbb{Q}$, proving the claim. □

The lemma resolves radical separation, not carrier compatibility. The first transport modulus $q^3(q-1)$ and the main carrier modulus $4q^2Y$ share the factor q^2. Rationality of a sum of quotients with these denominators does not imply their separate integrality. An attempted ten-variable equation must supply a new argument at exactly this point.

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